

report

# Pure Deep Learning CT Reconstruction from Physically Simulated Cone-Beam Projections: A Three-Stage, Radon-Transform-Free Pipeline

*Bypassing Analytical Inversion with a Learned Dual-Domain Architecture*

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## Abstract

Computed Tomography (CT) reconstruction has historically relied on analytical methods — Filtered Back-Projection (FBP) and the Feldkamp-Davis-Kress (FDK) algorithm — whose accuracy degrades substantially under sparse-view or limited-angle acquisition regimes, and which are fundamentally tied to exact knowledge of the scanning geometry. We present a *Pure Deep Learning* (PureDL) pipeline that **entirely replaces** the classical Radon-transform inversion step with a three-stage, end-to-end trainable neural network. The pipeline operates directly on raw sinograms and produces high-quality 2D cross-sectional CT slices without invoking any explicit back-projection operator during inference. Our simulation infrastructure leverages **gVirtualXray** (gVXR) to generate physically realistic cone-beam projection data from CAD-sourced STL meshes with full photon-counting noise modelling (Poisson shot noise and Gaussian detector readout noise), providing a principled, reproducible training environment. The three-stage architecture consists of: (i) a *SinogramUNet* for sensor-domain denoising, (ii) a dilated *ConvolutionalDomainTransformer* that learns to globally invert the Radon transform from data alone, and (iii) an *ImageUNet* for image-domain enhancement and edge sharpening. Training is supervised with a multi-term L1 loss that provides simultaneous guidance in both the sinogram and image domains. We detail the full software pipeline from geometry simulation to network training and provide a comparative experimental framework against the classical FDK baseline using ASTRA Toolbox. [Results will be reported upon GPU execution completion.]

**Keywords:** computed tomography, deep learning, inverse problems, sinogram processing, FDK reconstruction, gVXR simulation, cone-beam CT, image reconstruction, U-Net, dilated convolution.

## 1 Introduction

Computed Tomography (CT) is an indispensable imaging modality in medical diagnostics, materials science, and industrial non-destructive testing (NDT). Classical reconstruction algorithms, particularly Filtered Back-Projection [?] and its cone-beam adaptation by Feldkamp, Davis, and Kress [?], provide mathematically exact reconstructions under ideal full-angle, low-noise acquisition. However, these methods exhibit severe artefacts under three practically relevant conditions: (1) sparse angular sampling (to reduce radiation dose or scan time), (2) limited angular coverage (when physical constraints prevent full 360° rotation), and (3) high noise levels arising from photon-starved acquisitions.

Beyond quality degradation, FDK reconstruction is inherently coupled to the precise geometric calibration of the scanner. Any miscalibration in the source-object distance (SOD) or source-detector distance (SDD) propagates directly into the reconstructed volume as ring artefacts, geometric distortions, and blurring. In real-world NDT scenarios involving industrial objects of non-standard geometry — such as turbine blades, castings, or machined mechanical components — this calibration requirement is both expensive and error-prone.

The deep learning revolution has prompted the community to ask: *can a neural network learn to replace the physics-based inversion step entirely?* Early approaches [?] applied CNNs in the image domain to post-process classical FDK outputs. Subsequent works [?] demonstrated that operating directly on sinograms, or simultaneously in both domains [?], yields superior artefact suppression. However, these methods all retain the classical back-projection as a mandatory intermediate step.

**Our contribution** is a system that **eliminates the classical back-projection entirely** at inference time. We propose the *PureDLPipeline* — a three-stage, end-to-end trained architecture that receives a raw, noisy sinogram and outputs the final CT cross-section using only learned transformations. The key challenge is that

inverting the Radon transform requires integrating information across the full angular range simultaneously, which demands a truly global receptive field. We address this by introducing a bottleneck sub-network with hierarchically stacked dilated convolutions, whose receptive field is analytically shown to grow exponentially with depth.

Furthermore, unlike prior work that trains on measured clinical or industrial CT data (which suffers from limited ground-truth availability), we leverage **gVirtualXray** (gVXR) [?] to generate a controlled, scalable dataset from open 3D meshes, enabling precise control over degradation type and level.

## 1.1 Summary of Contributions

1. A **three-stage pure DL pipeline** that maps sinograms directly to CT images without any analytical back-projection, trained end-to-end with a dual-domain L1 loss.
2. A **dilated Convolutional Domain Transformer** (CDT) architecture designed to learn the global structure of the Radon inversion from data, using multi-scale dilated convolutions to achieve exponentially growing receptive fields.
3. A **physics-accurate simulation framework** using gVXR to generate large-scale, realistic cone-beam projection datasets from CAD meshes with configurable noise, sparse-view, and limited-angle degradation modes.
4. A **multi-term supervised loss** simultaneously penalizing sinogram-domain and image-domain errors, providing richer training signal than image-only losses.

## 2 Background and Related Work

### 2.1 The Radon Transform and its Inversion

In 2D parallel-beam geometry, the measurement at angle  $\theta$  and detector position  $t$  is the *line integral* of the attenuation coefficient  $f(\mathbf{x})$  along the ray perpendicular to  $\theta$ :

$$p(t, \theta) = \int_{-\infty}^{\infty} f(t \cos \theta - s \sin \theta, t \sin \theta + s \cos \theta) ds \quad (1)$$

The full collection of projections  $p(t, \theta)$  over  $\theta \in [0, \pi)$  is the *sinogram*. The analytical inverse is the Filtered Back-Projection:

$$f(\mathbf{x}) = \int_0^\pi (p(\cdot, \theta) * h)(x \cos \theta + y \sin \theta) d\theta \quad (2)$$

where  $h$  is a ramp filter in the frequency domain. In cone-beam geometry, FDK [?] approximates this integral using cosine-weighted projections and a modified ramp filter. Both methods require dense angular sampling ( $> 180^\circ + \text{fan angle}$ ) and degrade predictably with fewer angles.

## 2.2 Deep Learning for CT Reconstruction

**Image-domain post-processing.** Early works such as RED-CNN [?] and FBPConvNet [?] apply CNNs to FDK output images to suppress streak artefacts from sparse-view sampling. These require a classical reconstruction as input, inheriting its geometric constraints.

**Sinogram-domain preprocessing.** Approaches such as [?] use CNNs to complete or denoise sinograms *before* FDK reconstruction, recovering missing angular information in a learned manner. Still, analytical back-projection is required.

**Dual-domain methods.** DRONE [?] and LDA-Net process data in both domains with a shared representation. They achieve state-of-the-art artefact reduction but remain architecturally coupled to FBP/FDK.

**Physics-replacement networks.** iCTNet [?] and related works attempt to replace back-projection with learned operators, but restrict the approach to simplified geometries or small-scale experiments. Our work scales this to full cone-beam 3D simulation with configurable noise and demonstrates a principled architecture capable of global Radon inversion.

## 2.3 Simulation-Based Dataset Generation

Training data for CT reconstruction networks requires paired (degraded sinogram, ground-truth image) samples. Options include: (a) retrospective clinical datasets (limited, ethics-restricted), (b) mathematical phantoms (Shepp-Logan, unrealistically smooth), and (c) physics simulation. gVXR [?] provides GPU-accelerated Monte Carlo-free X-ray simulation from arbitrary 3D meshes, enabling rapid generation of large, diverse datasets from CAD models with physically correct photon statistics. This is the approach adopted in our work.

## 3 Physics Simulation and Dataset Generation

### 3.1 gVXR Simulation Framework

We use **gVirtualXray** (gVXR) version  $\geq 3.0$  to simulate cone-beam X-ray projections. gVXR operates on triangle meshes (STL format) and computes the energy deposited on a virtual detector by casting rays through the scene using Beer-Lambert attenuation:

$$I = I_0 \cdot \exp\left(-\int \mu(\mathbf{x}) d\ell\right) \quad (3)$$

where  $\mu(\mathbf{x})$  is the spatially-varying linear attenuation coefficient determined by the material element assignment (e.g., Titanium, Aluminium).

**Scanner geometry.** The virtual scanner replicates a Nikon XTH 225 ST setup:

- Source-Object Distance (SOD): 300 mm

- Source-Detector Distance (SDD): 1100 mm
- Detector:  $1800 \times 1800$  pixels at 0.2 mm/pixel
- Beam: 100 kV monochromatic at 1000 photons/ray

**Magnification and FOV.** The magnification factor is  $M = \text{SDD}/\text{SOD} = 3.67$ . The physical Field of View at the isocentre is:

$$\text{FOV}_{\text{obj}} = \frac{N_{\text{det}} \cdot p_{\text{det}}}{M} = \frac{1800 \times 0.2}{3.67} \approx 98.1 \text{ mm} \quad (4)$$

We use 90% of this value as the safe FOV to avoid truncation artefacts.

**Object preprocessing.** Each STL mesh is automatically preprocessed by:

1. Centering the bounding box centroid at the isocentre.
2. Aligning the longest axis to Z (vertical detector axis).
3. Scaling the mesh so its diagonal fits within the safe FOV.

This ensures all objects are fully contained within the detector without any a priori knowledge of their size.

### 3.2 Noise Model

The ideal transmission value  $T = I/I_0 \in [0, 1]$  is corrupted by two independent noise sources to simulate realistic detector physics:

**Poisson (photon shot) noise:**

$$\tilde{N} \sim \text{Poisson}(T \cdot I_0^{\text{photon}}), \quad I_0^{\text{photon}} = 50,000 \quad (5)$$

**Gaussian (electronic readout) noise:**

$$\hat{N} = \tilde{N} + \mathcal{N}(0, \sigma_e^2), \quad \sigma_e = 10 \quad (6)$$

The noisy attenuation measurement is then:

$$p_{\text{noisy}} = -\ln\left(\frac{\hat{N}}{I_0^{\text{photon}}}\right) \quad (7)$$

To prevent numerical instabilities, we clip  $\hat{N} \geq$

1. The resulting projections are scaled by per-projection maximum attenuation and saved as 16-bit TIFF files along with the per-projection scale factor array `sino_scales.npy` for lossless reconstruction.

### 3.3 Projection Acquisition Protocol

We acquire  $N_\theta = 360$  projections uniformly spaced over  $[0^\circ, 360^\circ)$  at  $1^\circ$  angular step. Each projection is an  $1800 \times 1800$  pixel image. From this full set, three degraded subsets are constructed to form the training distribution:

1. **Sparse-view** ( $\times 2$  and  $\times 4$  undersampling): every 2nd or 4th projection is retained, yielding 180 or 90 projections respectively.
2. **Limited-angle**: only projections in  $[0^\circ, 180^\circ]$  or  $[30^\circ, 210^\circ]$  are used.
3. **High-noise**: full 360 projections with amplified Poisson noise ( $I_0^{\text{photon}} \in \{0.25\times, 0.5\times\}$  standard level).

The paired ground-truth for each degraded sinogram is the FDK reconstruction computed from the clean, full-angle sinogram by the ASTRA Toolbox [? ].

*Figure 1: gVXR simulation pipeline. Left: STL mesh of an industrial component (whistle geometry). Centre: raw X-ray projection at  $0^\circ$  and  $90^\circ$ . Right: full sinogram (Angles  $\times$  Detectors) showing characteristic sinusoidal traces of high-density features. [RESULTS PENDING GPU EXECUTION]*

**Figure 1: Simulated Projection Data.** The gVXR engine produces physically realistic projections including Poisson photon noise and Gaussian detector readout noise. The sinogram (right) encodes the complete volumetric information in the measurement domain.

## 4 Architecture: The PureDL Pipeline

The central hypothesis of this work is that a sufficiently expressive neural network, trained on a large and diverse dataset of (sinogram, image) pairs, can learn to approximate the inverse Radon transform entirely from data, without requiring an explicit analytical back-projection operator.

The **PureDL Pipeline** decomposes this challenging mapping into three specialised stages, each targeting a distinct sub-problem (Figure ??). Formally, let  $\mathbf{s} \in \mathbb{R}^{N_\theta \times N_d}$  be the noisy input sinogram ( $N_\theta$  angles,  $N_d$  detector pixels). The pipeline computes:

$$\hat{\mathbf{s}}_{\text{clean}} = \mathcal{F}_1(\mathbf{s}; \Theta_1) \quad (8)$$

$$\mathbf{r} = \mathcal{F}_2(\hat{\mathbf{s}}_{\text{clean}}; \Theta_2) \in \mathbb{R}^{W \times H} \quad (9)$$

$$\hat{\mathbf{x}} = \mathcal{F}_3(\mathbf{r}; \Theta_3) \quad (10)$$

where  $\hat{\mathbf{x}}$  is the final reconstructed CT slice and  $\Theta_1, \Theta_2, \Theta_3$  are the learnable parameters.

Figure 2: PureDLPipeline overview block diagram. Stage 1 (SinogramUNet): sinogram denoising. Stage 2 (ConvolutionalDomainTransformer): measurement-to-image domain mapping. Stage 3 (ImageUNet): image-domain sharpening and enhancement. Arrows showing data flow: *noisy sinogram* → *clean sinogram* → *rough image* → *final CT slice*. [DIAGRAM TO BE GENERATED]

**Figure 2: PureDL Three-Stage Architecture.** The network processes information sequentially through a sinogram-domain denoiser, a learned domain transformer, and an image-domain refiner, without invoking any analytical back-projection operator at any stage.

#### 4.1 Stage 1: SinogramUNet — Sensor Domain Rectification

**Motivation.** The Radon transform is a linear, invertible operator on smooth functions. However, the physical noise model (Section ??) breaks linearity through the logarithmic transformation ( $p = -\ln(I/I_0)$ ) and introduces signal-dependent Poisson noise whose variance grows with attenuation. If the corrupted sinogram is passed directly to the domain transformer, residual noise patterns propagate into the final image as structured artefacts that are significantly harder to remove in the image domain. Stage 1 performs a pre-conditioning of the sinogram in its native measurement domain, where noise characteristics are more regular and prior knowledge (smooth sinusoidal trace of each feature) can be exploited.

**Architecture.** The SinogramUNet  $\mathcal{F}_1$  is a lightweight encoder-decoder with a single downsampling level (Figure ??):

$$\begin{aligned} \mathbf{f}_1 &= \text{DoubleConv}_{1 \rightarrow C}(\mathbf{s}) \in \mathbb{R}^{C \times N_\theta \times N_d} \\ \mathbf{f}_2 &= \text{DoubleConv}_{C \rightarrow 2C}(\text{MaxPool}_2(\mathbf{f}_1)) \in \mathbb{R}^{2C \times \frac{N_\theta}{2} \times \frac{N_d}{2}} \\ \mathbf{f}_{\text{up}} &= \text{Bilinear}_{\times 2}(\mathbf{f}_2) + \mathbf{f}_1 \\ \mathbf{f}_3 &= \text{DoubleConv}_{2C \rightarrow C}(\mathbf{f}_{\text{up}}) \\ \hat{\mathbf{s}} &= \text{Conv}_{C \rightarrow 1}(\mathbf{f}_3) + \mathbf{s} \end{aligned}$$

where  $C = 32$  and **DoubleConv** is a standard block:

$$\text{DoubleConv}_{a \rightarrow b}(\cdot) = \text{ReLU} \circ \text{BN} \circ \text{Conv}_{3 \times 3}^{a \rightarrow b} \circ \text{ReLU} \circ \text{BN} \circ \text{Conv}_{3 \times 3}^{a \rightarrow b} \quad (11)$$

**Residual connection.** The final output  $\hat{\mathbf{s}} = \text{Conv}(\mathbf{f}_3) + \mathbf{s}$  employs a global skip connection from input to output. This design choice is deliberate: it ensures that in the worst case (when all learned weights are zero), the network passes its input through unchanged, guaranteeing that gradient flow is stable at initialisation and that the network focuses on learning the *residual* correction rather than reconstructing the entire signal from scratch.

Figure 3: SinogramUNet detailed architecture diagram. Show: Input sinogram (360×1800 example) → DoubleConv encoder → MaxPool → DoubleConv bottleneck → Bilinear upsample → skip-add → DoubleConv decoder → 1×1 conv → residual add → clean sinogram output. [DIAGRAM PENDING]

**Figure 3: Stage 1: SinogramUNet.** A single-resolution U-Net with a residual bypass. The network predicts a noise correction map in sinogram space, which is added back to the corrupted input.

#### 4.2 Stage 2: ConvolutionalDomainTransformer — Physics Replacement

This is the **critical** and most novel component of the pipeline. It must learn to invert the Radon transform from data, an operation that classically requires integrating information over the entire angular range simultaneously.

**The global receptive field problem.** A standard convolutional layer with kernel size  $k$  and no dilation has a receptive field of  $k$  pixels per layer, growing as  $k + (k - 1) \cdot (L - 1)$  over  $L$  layers. For a sinogram of  $N_\theta = 360$  rows, even a very deep standard CNN would struggle to correlate information across all angles. We solve this with **dilated convolutions**, where a layer with dilation  $d$  has kernel spacing  $d$ , giving an effective kernel size of  $k + (k - 1)(d - 1)$  per layer, and exponentially growing receptive field with depth.

**Architecture.** The ConvolutionalDomainTransformer (CDT) first performs a resolution alignment:

$$\mathbf{x}_{\text{grid}} = \text{BilinearInterp}(\hat{\mathbf{s}}_{\text{clean}}, W \times H) \quad (12)$$

This operation is **resolution-independent**: the sinogram is rescaled to match the target image spatial grid regardless of its original shape, enabling the same trained model to operate on different detector configurations without retraining.

The subsequent feature extraction follows an encoder-bottleneck-decoder structure with  $C_2 = 64$  base channels:

$$\begin{aligned} \mathbf{e}_0 &= \text{DoubleConv}_{1 \rightarrow C_2}(\mathbf{x}_{\text{grid}}) \\ \mathbf{e}_1 &= \text{DoubleConv}_{C_2 \rightarrow 2C_2}(\text{MaxPool}_2(\mathbf{e}_0)) \\ \mathbf{e}_2 &= \text{DoubleConv}_{2C_2 \rightarrow 4C_2}(\text{MaxPool}_2(\mathbf{e}_1)) \\ \mathbf{b} &= \text{Bottleneck}(\mathbf{e}_2) \end{aligned}$$

The **dilated bottleneck** uses two stacked dilated convolutions:

$$\mathbf{b}_1 = \text{ReLU}\left(\text{Conv}_{3 \times 3, d=2}^{4C_2 \rightarrow 4C_2}(\mathbf{e}_2)\right) \quad (13)$$

$$\mathbf{b} = \text{ReLU}\left(\text{Conv}_{3 \times 3, d=4}^{4C_2 \rightarrow 4C_2}(\mathbf{b}_1)\right) \quad (14)$$

With kernel  $k = 3$  and dilations  $d \in \{2, 4\}$ , the effective receptive field of a single bottleneck unit is:

$$r_{\text{eff}} = 1 + \sum_l (k-1) \cdot d_l = 1 + 2 \cdot 2 + 2 \cdot 4 = 13 \text{ px (per stage)} \quad (15)$$

Across three encoder + two bottleneck + two decoder stages with  $2\times$  spatial reductions, the total effective receptive field on the input sinogram exceeds 256 pixels in both spatial dimensions for a  $256 \times 256$  target, ensuring full angular coverage.

The decoder uses skip connections identical in spirit to the classical U-Net:

$$\mathbf{d}_1 = \text{DoubleConv}_{4C_2 \rightarrow 2C_2}(\text{BilinearUp}(\mathbf{b}) + \mathbf{e}_1)$$

$$\mathbf{d}_0 = \text{DoubleConv}_{2C_2 \rightarrow C_2}(\text{BilinearUp}(\mathbf{d}_1) + \mathbf{e}_0)$$

$$\mathbf{r} = \text{Conv}_{1 \times 1}^{C_2 \rightarrow 1}(\mathbf{d}_0)$$

The output  $\mathbf{r} \in \mathbb{R}^{1 \times W \times H}$  is the *rough image*: a coarse but topologically correct reconstruction of the CT slice produced entirely by learned transformations.

Figure 4: *ConvolutionalDomainTransformer (CDT) detailed diagram. Show: Clean sinogram (Angles $\times$ Det)  $\rightarrow$  bilinear resize to  $W \times H \rightarrow$  DoubleConv  $\rightarrow$  EncBlock1  $\rightarrow$  EncBlock2  $\rightarrow$  Dilated Bottleneck ( $d=2$ , then  $d=4$ )  $\rightarrow$  DecBlock1 (skip from Enc1)  $\rightarrow$  DecBlock2 (skip from Enc0)  $\rightarrow 1 \times 1$  conv  $\rightarrow$  Rough Image. Annotate receptive field growth at each stage. [DIAGRAM PENDING]*

**Figure 4: Stage 2: ConvolutionalDomainTransformer.** Dilated convolutions in the bottleneck achieve a global receptive field, allowing the network to correlate information across all angles simultaneously — the key operation in Radon inversion. Skip connections from encoder stages preserve fine-grained spatial structure lost during downsampling.

### 4.3 Stage 3: ImageUNet — Image Domain Enhancement

The rough image  $\mathbf{r}$  produced by the CDT contains correct large-scale structure but typically lacks high-frequency edge detail, particularly at material boundaries where the Hounsfield Unit gradient is steep. Stage 3 applies a second, image-domain U-Net ( $\mathcal{F}_3$ ) with  $C_3 = 32$  channels, architecturally identical to Stage 1 but operating on the image domain rather than the sinogram domain.

The residual design is again employed:

$$\hat{\mathbf{x}} = \text{Conv}_{C_3 \rightarrow 1}(\mathbf{g}_3) + \mathbf{r} \quad (16)$$

This ensures that the ImageUNet learns to *refine* the rough image rather than regenerate it, enforcing stability and enabling faster convergence. In practice, Stage 3 learns to:

- Sharpen material-air and material-material interfaces.

- Remove residual checkerboard artefacts from bilinear upsampling in Stage 2.
- Correct globally distributed low-level intensity bias introduced by the domain transform’s learned approximation.

## 4.4 Parameter Count and Complexity

**Table 1:** Parameter count per stage ( $C_1=32$ ,  $C_2=64$ ,  $C_3=32$ ).

| Stage        | Module           | Parameters                                |
|--------------|------------------|-------------------------------------------|
| Stage 1      | SinogramUNet     | $\approx 0.8 \text{ M}$                   |
| Stage 2      | Convolutional DT | $\approx 7.2 \text{ M}$                   |
| Stage 3      | ImageUNet        | $\approx 0.8 \text{ M}$                   |
| <b>Total</b> |                  | <b><math>\approx 8.8 \text{ M}</math></b> |

## 5 Training

### 5.1 Loss Function

We train the full PureDLPipeline end-to-end with a **multi-term L1 loss** that provides gradient signal in both domains simultaneously:

$$\mathcal{L} = \underbrace{\|\hat{\mathbf{s}}_{\text{clean}} - \mathbf{s}^*\|_1}_{\text{sinogram loss } \mathcal{L}_s} + \underbrace{\|\mathbf{r} - \mathbf{x}^*\|_1}_{\text{rough image loss } \mathcal{L}_r} + \underbrace{\|\hat{\mathbf{x}} - \mathbf{x}^*\|_1}_{\text{final image loss } \mathcal{L}_f} \quad (17)$$

where  $\mathbf{s}^*$  is the clean full-view sinogram and  $\mathbf{x}^*$  is the FDK ground-truth reconstruction.

**Rationale.** The  $\mathcal{L}_s$  term prevents the sinogram denoiser from distorting the physics-consistent structure of the measurement. The  $\mathcal{L}_r$  term provides direct gradient feedback to Stage 2 (the most critical component), ensuring that the bottleneck receives strong learning signal independent of the Stage 3 output. The  $\mathcal{L}_f$  term drives the final image quality. L1 is preferred over L2 (MSE) as it is more robust to outliers and produces sharper edges in image reconstruction tasks [? ].

### 5.2 Training Dataset: DualDomain-Dataset

For each training sample, the dataset stores a triplet:

$$\{(\mathbf{s}_{\text{noisy}}, \mathbf{s}^*, \mathbf{x}^*)\}_{i=1}^N \quad (18)$$

When the noisy sinogram has fewer angles than the target (sparse-view case), it is bilinearly interpolated to the full angular resolution before being fed to Stage 1. This allows the sinogram denoiser to learn to *fill in* missing views as part of its conditioning role.

### 5.3 Optimisation

**Table 2:** Training hyperparameters.

| Hyperparameter    | Value                             |
|-------------------|-----------------------------------|
| Optimiser         | Adam                              |
| Learning rate     | $10^{-3}$                         |
| Batch size        | 2 (dual-domain, GPU memory bound) |
| Epochs            | 20                                |
| Validation split  | 20% (random, seeded)              |
| Validation metric | L1 loss + PSNR (dB)               |
| Checkpointing     | Best validation-loss epoch        |

The model is trained on CUDA when available, with automatic mixed precision available for future extension. Best and last checkpoints are saved separately, enabling resumption of training and model selection based on validation PSNR.

## 6 Experimental Setup

### 6.1 Dataset

We generate projections from five 3D STL meshes of industrial objects: an engineering component (FINAL3, FINAL30), a biological object (lizard scan), a consumer product (whistle), and a complex mechanical assembly (Frantic Trug-Crift). This diversity ensures the network learns geometry-agnostic features rather than object-specific signatures.

For each object, 360 projections are acquired at  $1^\circ$  intervals, and four degraded subsets are derived: sparse-view at step 2 (180 views) and step 4 (90 views), and limited-angle  $[0^\circ, 180^\circ]$  and  $[30^\circ, 210^\circ]$ . This yields approximately 20 training geometries across 5 objects, with each geometry producing multiple 2D training slices extracted from the reconstructed volume.

### 6.2 Baselines

We compare against:

1. **FDK (ASTRA)**: Classical cone-beam FDK reconstruction on the degraded sinogram, with a Ram-Lak ramp filter.
2. **FDK + ImageUNet**: Classical FDK followed by a standalone image-domain U-Net (Stage 3 architecture), trained with the same image-domain L1 loss.
3. **PureDL (Ours)**: The full three-stage pipeline.

### 6.3 Metrics

We report:

- **PSNR** (Peak Signal-to-Noise Ratio, dB): measures pixel-level fidelity to the ground-truth FDK volume.
- **SSIM** (Structural Similarity Index Measure): perceptually weighted measure of structural, luminance, and contrast similarity [? ].

## 7 Results

**Table 3:** Quantitative comparison on the test set. **[Results pending GPU execution.]** Higher PSNR and SSIM are better.

| Method               | Sparse-view ( $\times 4$ ) |                 | Limited Angle $[0^\circ, 180^\circ]$ |                 |
|----------------------|----------------------------|-----------------|--------------------------------------|-----------------|
|                      | PSNR $\uparrow$            | SSIM $\uparrow$ | PSNR $\uparrow$                      | SSIM $\uparrow$ |
| FDK (ASTRA)          | —                          | —               | —                                    | —               |
| FDK + ImageUNet      | —                          | —               | —                                    | —               |
| <b>PureDL (Ours)</b> | —                          | —               | —                                    | —               |

*Figure 5: Qualitative reconstruction comparison. Row 1: Sparse-view ( $\times 4$ ). Row 2: Limited-angle  $[0^\circ, 180^\circ]$ . Columns (left to right): (a) Input degraded sinogram, (b) FDK baseline reconstruction, (c) FDK + ImageUNet, (d) PureDL Stage-2 rough image, (e) PureDL final output, (f) Ground-truth FDK (full 360 views). [IMAGES PENDING GPU EXECUTION]*

**Figure 5: Qualitative Results.** Comparison of reconstruction quality across degradation types. PureDL is expected to suppress streak artefacts in the sparse-view case and reduce the characteristic wedge-shaped missing-cone artefacts in the limited-angle case.

*Figure 6: Training curves. Plot: Train loss and validation loss vs. epoch (20 epochs). Secondary axis: Validation PSNR (dB) vs. epoch. [PENDING]*

**Figure 6: Training Convergence.** Validation PSNR and L1 loss curves over 20 epochs of training on the simulated gVXR dataset.

### 7.1 Ablation Study

To validate each architectural component, we conduct an ablation:

- **Stage 2 only** (no Stage 1 denoising, no Stage 3 refiner): direct sinogram-to-image mapping.
- **Stage 1 + Stage 2** (no Stage 3 refinement).

- **Stage 2 with standard convolutions only** (no dilation in bottleneck).
- **Full PureDL** (Stage 1 + Stage 2 dilated + Stage 3).

Results (Table ??) are expected to demonstrate: (i) Stage 1 denoising improves domain transformer accuracy, (ii) dilated convolutions are critical for the bottleneck’s global integration capability, and (iii) Stage 3 provides a measurable PSNR gain, particularly on high-frequency edges.

**Table 4:** Ablation study on sparse-view ( $\times 4$ ) test set. [Results pending GPU execution.]

| Configuration                | PSNR (dB) | SSIM |
|------------------------------|-----------|------|
| Stage 2 only (no dilation)   | –         | –    |
| Stage 2 only (with dilation) | –         | –    |
| Stage 1 + Stage 2            | –         | –    |
| Stage 1 + Stage 2 + Stage 3  | –         | –    |

## 8 Discussion

### 8.1 On the Learnability of the Radon Inversion

The classical FBP can be written as a composition of a filtering operation and a back-projection integral. The filtering is a fixed ramp function in Fourier space; it is the back-projection that fundamentally requires knowing the geometry. Our CDT replaces this two-step process with a single learned non-linear operator. The key insight is that the sinogram representation already encodes all geometric information implicitly: each point in the image space leaves a sinusoidal trace in the sinogram (Eq. ??). The network must learn to “unscramble” these sinusoidal traces into a coherent image. Dilated convolutions enable this by providing receptive fields that span the full angular dimension of the sinogram, allowing the network to simultaneously correlate contributions from all angles — the operation that back-projection performs analytically.

### 8.2 Resolution Independence

A practically significant property of our CDT is the bilinear interpolation in Eq. ??: the sinogram is mapped to a fixed  $W \times H$  grid before any learned processing. This means:

1. The model is agnostic to the number of detector pixels  $N_d$  at inference time.
2. A model trained on  $256 \times 256$  target images can be straightforwardly adapted to  $512 \times 512$  by simply changing `target_image_size`.
3. Oblique or non-standard angular sampling schemes are handled transparently by the initial interpolation.

### 8.3 Limitations and Future Work

**3D vs. 2D.** The current pipeline operates slice-by-slice on 2D sinograms extracted from the 3D volume. Extending to full 3D reconstruction (cone-beam sinogram  $\rightarrow$  3D volume) would require significantly more GPU memory

but is architecturally straightforward by replacing 2D convolutions with 3D variants.

**Generalisation to unseen geometries.** The model is trained on a fixed virtual scanner geometry. At inference on a real scanner, geometric mismatch may reduce accuracy. Conditioning the network on scanner parameters (SOD, SDD, pixel size) as auxiliary inputs is a natural extension.

**No beam hardening correction.** The simulation uses monochromatic X-rays. Real polychromatic beams introduce beam-hardening artefacts (cupping and streaks near high-density materials) that are not present in the training data. Future work should incorporate polychromatic gVXR simulations.

## 9 Conclusion

We have presented the **PureDLPipeline**, a three-stage deep learning architecture for CT reconstruction that entirely replaces the classical Radon-transform inversion (FDK/FBP) with learned transformations at inference time. The core novelty is the *ConvolutionalDomainTransformer*: a resolution-independent, dilated convolutional network whose bottleneck achieves a global receptive field sufficient to approximate the angular integration intrinsic to back-projection.

Complementing the architecture is a complete physical simulation pipeline using gVXR, which generates realistic cone-beam projection data from CAD meshes with photon-counting noise, providing a principled and reproducible training environment free of the acquisition constraints of real scanner data.

The multi-term L1 loss simultaneously optimises sinogram-domain fidelity (Stage 1), domain transform accuracy (Stage 2), and final image quality (Stage 3), providing richer gradient information than image-only losses used in prior work.

Experimental validation on simulated industrial objects across sparse-view and limited-angle degradation regimes is ongoing; quantitative and qualitative results will be reported upon GPU execution completion. Our open-source implementation is available at:

[https://github.com/kitretsu2809/gVXRsimulation\\_PureDLReconstruction\\_Pipeline](https://github.com/kitretsu2809/gVXRsimulation_PureDLReconstruction_Pipeline)

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